

ENCN205

Applied Data Analysis for Civil and Natural Systems

LECTURE 4

Coordinate Reference Systems & Projections

Week 2 | L4

University of Canterbury | Te Whare Wānanga o Waitaha

Dr Alberto Ardid

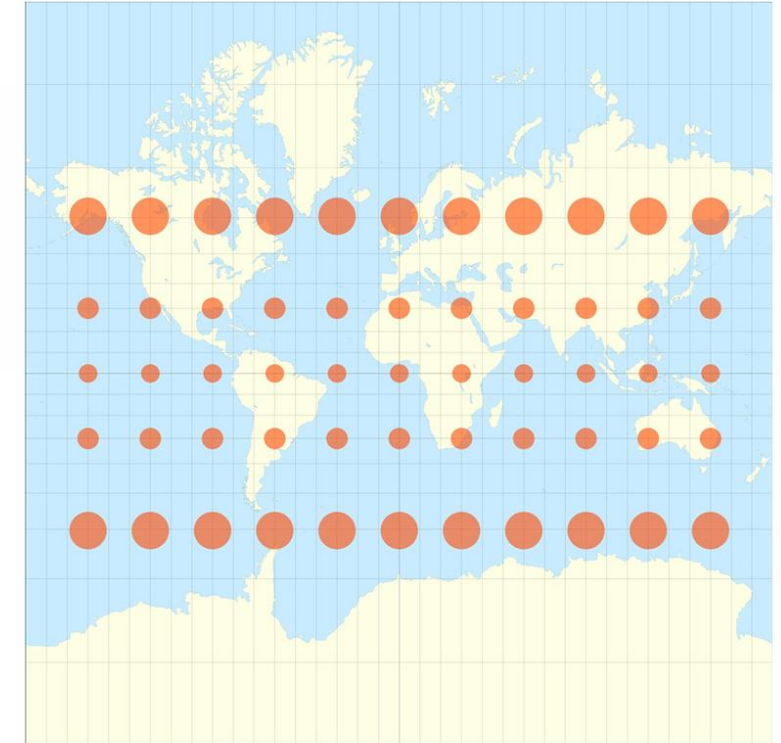
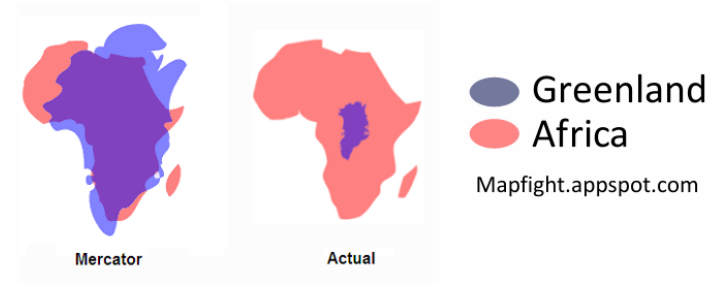
Lecturer / Assistant Professor
Department of Civil and Environmental Engineering
University of Canterbury, New Zealand

E319 |
alberto.ardid@canterbury.ac.nz



The Greenland Problem

On a standard world map, Greenland looks as big as Africa.



<https://www.vox.com/world/2016/12/2/13817712/map-projection-mercator-globe>

But Africa is 14 times larger than Greenland.

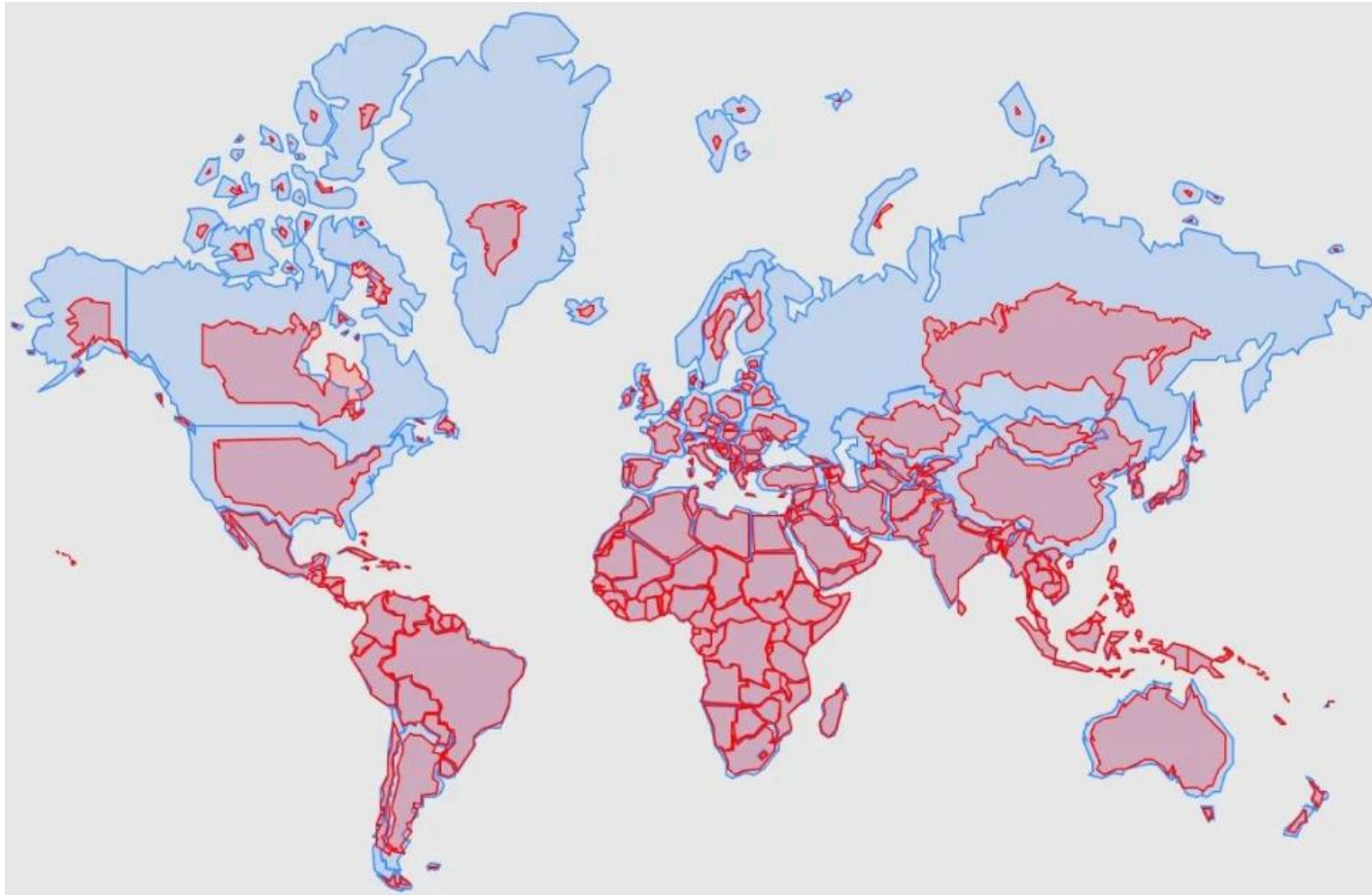
Africa: 30.4 million km²

Greenland: 2.2 million km²

This distortion is caused by the Mercator projection — a choice made in 1569 that still shapes how we see the world.

Every map projection distorts something. The question is: what can you afford to distort?

The Greenland Problem — this week, at the United Nations



Blue: countries as Mercator draws them. Red: their true relative size. Source: The Guardian, 4 Sep 2026

164 to 1: the UN votes to 'Correct the Map'

UN General Assembly, 4 September 2026. Resolution led by Togo and the African Union member states; 164 in favour, 1 against, 6 abstentions.

Recommends equal-area projections, especially Equal Earth (2018), whenever a map is used to show the relative size of countries and continents.

Non-binding, and Mercator remains 'fully suited' for maritime and aerial navigation. The right tool for the job, not a banned tool.

“A fair map does not change the geography of the world, it changes how we see the world.”

— Robert Dussey, Foreign Minister of Togo

theguardian.com/world/2026/sep/04/un-vote-world-map-mercator-equal-earth-africa

Why Engineers Must Care About CRS

Wrong CRS = wrong distances, wrong areas, wrong decisions

If you calculate a buffer distance in degrees instead of metres, your result is meaningless

If two layers use different CRS and you overlay them, features appear in the wrong locations

If you measure area using a projection that distorts area, your cost estimate is wrong

If you mix up NZGD2000 and NZTM2000, your coordinates shift by thousands of units

CRS is invisible but critical -- like units in physics. Get it wrong and nothing downstream is valid.

Earth's Three Surfaces

Three different surfaces are used to model, measure, and understand our planet.

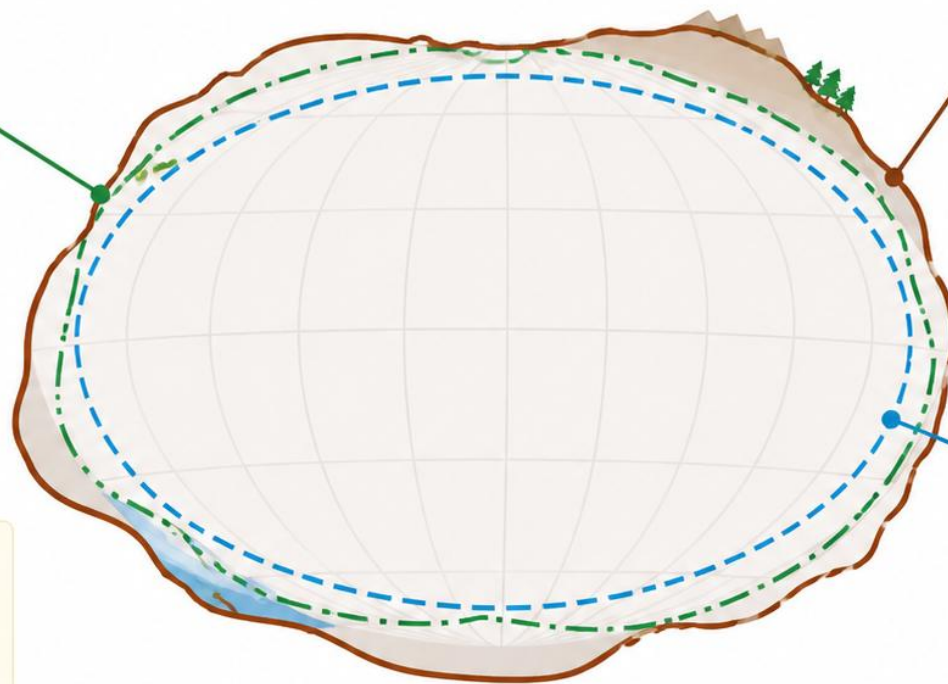
Geoid (gravity surface / mean sea level)

An equipotential surface that best represents mean sea level. It undulates due to variations in Earth's gravity field.



Key idea

The differences between these surfaces are small (tens of meters) but important for accurate mapping and positioning.



Earth's actual (physical) surface

The real, irregular surface we live on — includes mountains, valleys, oceans, and everything in between.

Ellipsoid (smooth mathematical approximation)

A simple, smooth geometric surface (usually an oblate spheroid) used as a reference for mapping, positioning, and calculations.

Surface	What it represents	How it's defined	Typical use	Relative to others
 Ellipsoid (mathematical)	A smooth, regular model of the Earth.	Mathematically defined (e.g., WGS 84).	GPS, mapping, GIS, and calculations.	 Smooth reference surface.
 Geoid (gravity surface)	Mean sea level extended under the continents.	Based on gravity measurements.	Height systems (e.g., orthometric height), oceanography.	 Follows gravity; undulates above and below ellipsoid.
 Earth's actual (physical surface)	The real, irregular surface of the Earth.	Observed and measured (topography, bathymetry).	Visualization, engineering, environmental studies.	 Above both; varies greatly in height.



Note: The separation between the geoid and the ellipsoid is called the geoid undulation (N), which ranges from about -100 m to +100 m globally.

Datum = ellipsoid + its position relative to the Earth. Different datums = different coordinate values for the same point.

Geographic CRS — Latitude & Longitude

Angles on a curved surface

Latitude: angle north/south of the equator (-90° to 90°)

Longitude: angle east/west of the Prime Meridian (-180° to 180°)

WGS84 (EPSG:4326)

Global standard — used by GPS

NZGD2000 (EPSG:4167)

NZ realisation — practically identical to WGS84

Differences are sub-metre (negligible for most work)

Christchurch example:

Latitude: -43.5321°

Longitude: 172.6362°

These are ANGLES, not distances.

1° of latitude ≈ 111 km everywhere

1° of longitude ≈ 80 km at 44°S

(varies with latitude!)

The Problem with Degrees

Degrees do not have a consistent physical meaning

1° of longitude at the equator = 111.3 km

1° of longitude at 45°S (Canterbury) = 78.8 km

1° of longitude at 90° (pole) = 0 km

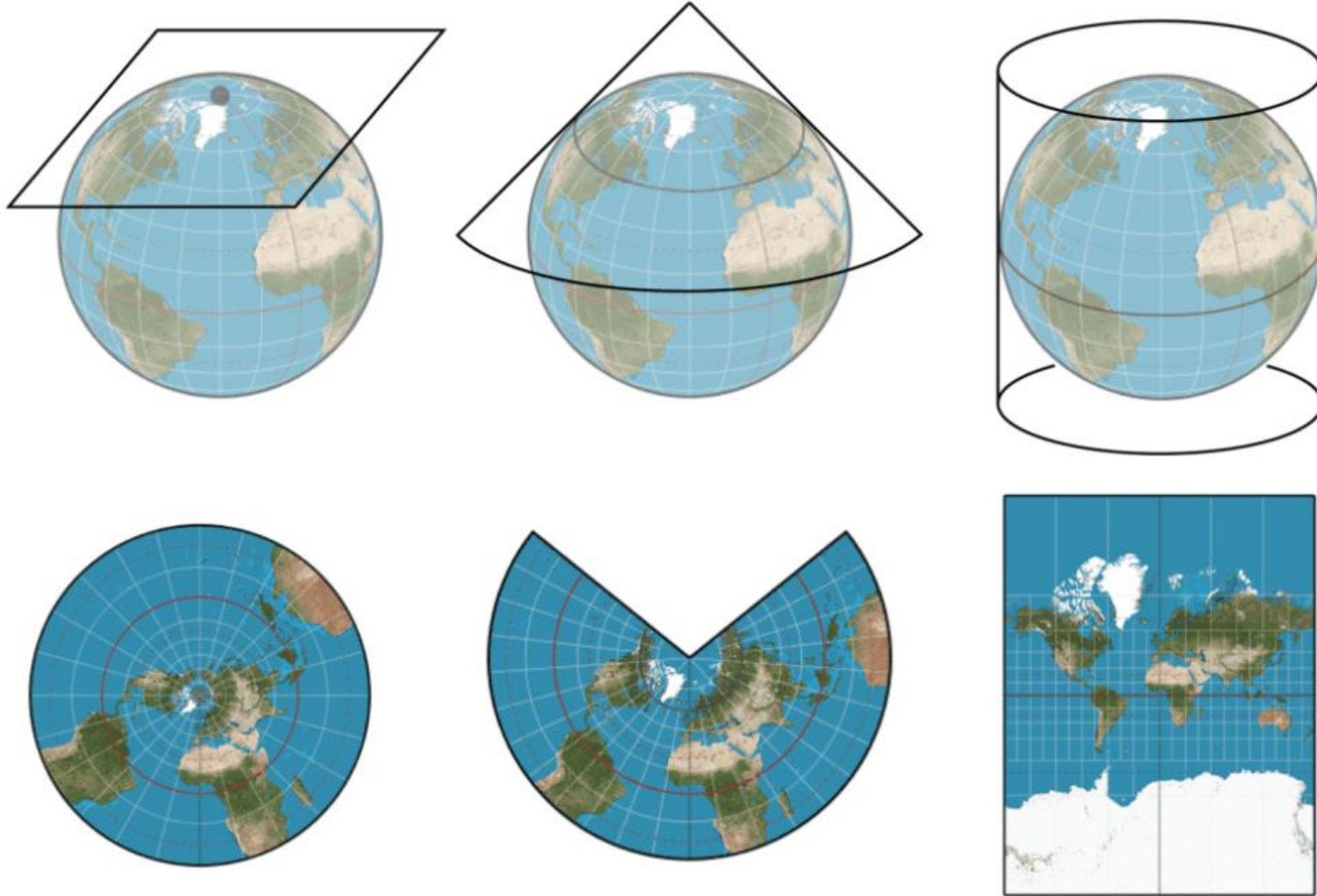
So what does a '500 m buffer' mean in degrees?

It depends on where you are — which makes it useless for engineering.

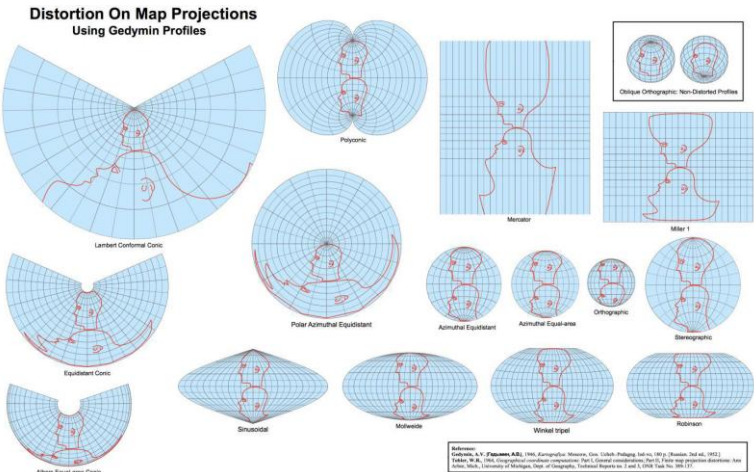
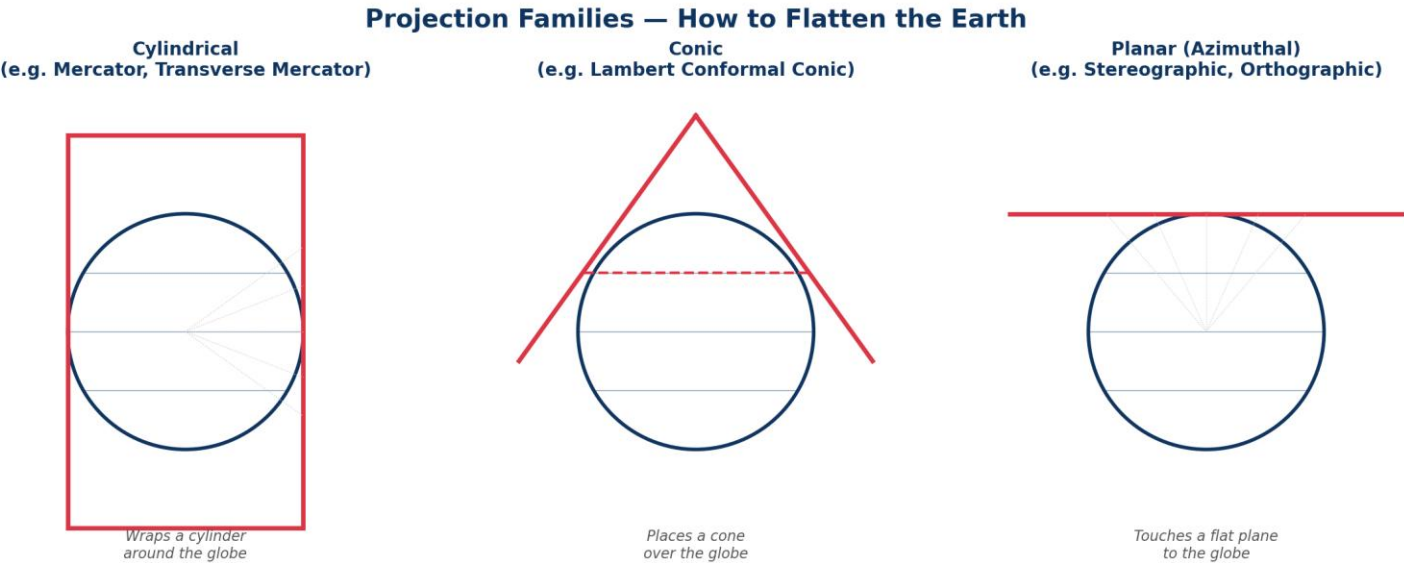
Solution: project to a CRS with metre-based coordinates.

In NZ: use NZTM2000 (EPSG:2193) for all distance and area calculations.

Projected CRS — Flattening the Earth



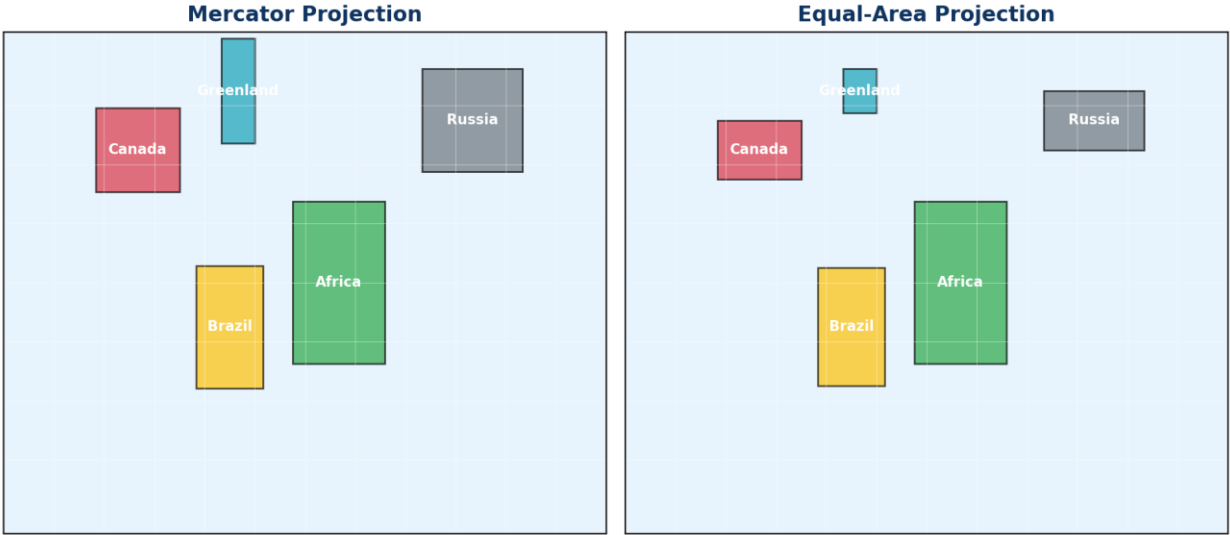
Projected CRS — Flattening the Earth



A projection transforms 3D coordinates (lat/lon on a sphere) into 2D coordinates (x, y on a flat surface).
Each family has different distortion characteristics — choose based on your analysis needs.

Distortion Trade-Offs

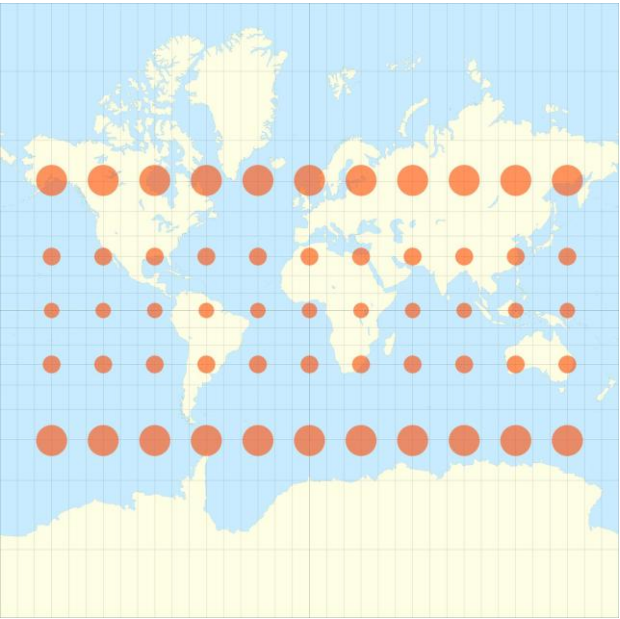
Projection Distortion — No Projection Is Perfect



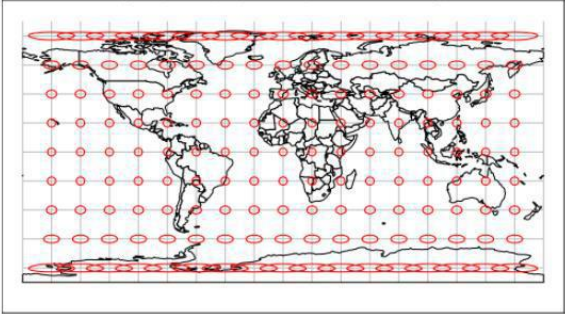
Preserves shape, distorts area

Preserves area, distorts shape

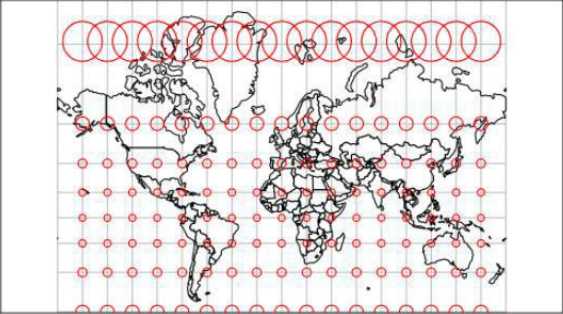
Greenland appears as large as Africa in Mercator — in reality Africa is 14x bigger!



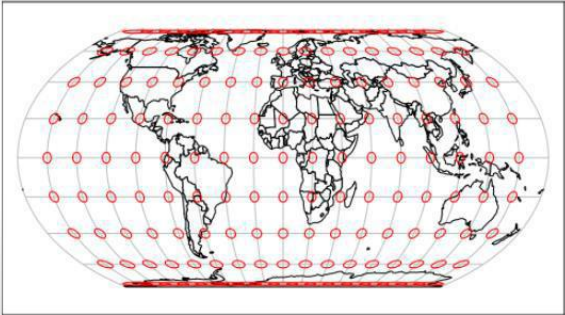
Equidistant Cylindrical / Plate Carree
(sf::plot, sp::plot, ggplot2::geom_sf)



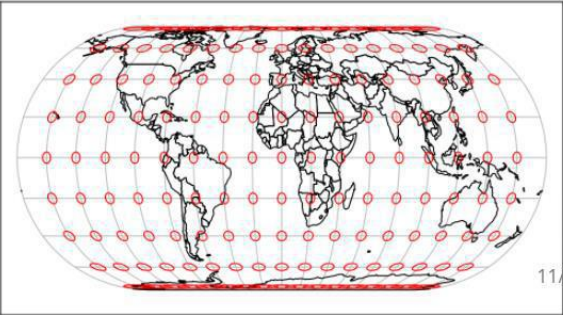
Web Mercator
(leaflet / mapview)



Equal Earth



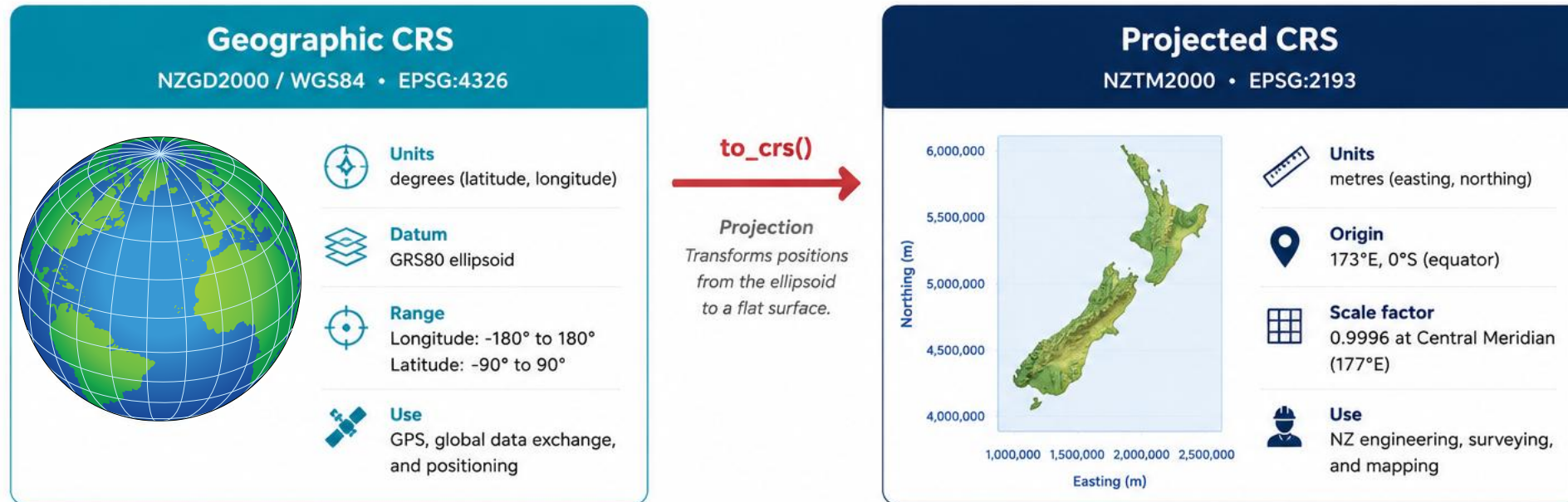
Eckhart-IV (tmap)



Every projection preserves some properties and distorts others: area, shape, distance, or direction.

New Zealand Coordinate Reference Systems

Two main types of CRS are used in New Zealand — geographic for global positioning, and projected for accurate measuring and mapping.



Geographic CRS

- 3D, matches the Earth's curvature
- Uses latitude & longitude
- Best for global data and positioning



Conversion

Projection converts ellipsoidal coordinates (lat/lon) to planar coordinates (x/y).



Projected CRS

- 2D, flat map representation
- Uses easting & northing (metres)
- Best for accurate distance, area, and engineering work



Key takeaway

Use a geographic CRS (NZGD2000 / WGS84) for global positioning and data exchange.
Use a projected CRS (NZTM2000) for accurate measurement and mapping within New Zealand.



Meridional circuits (local projections)

New Zealand also uses meridional circuits based on NZGD2000 for local surveying and cadastral work.
Examples: NZGD2000 / NZTM2000, NZGD2000 / Otago 2000, NZGD2000 / Southland 2000, etc.



For this course: EPSG:4326 (geographic) and EPSG:2193 (projected NZTM). Know both codes.

NZTM2000 — The Standard for NZ Engineering

New Zealand Transverse Mercator 2000

EPSG code: 2193

Units: metres

Central meridian: 173°E

Origin latitude: equator (0°)

False easting: 1,600,000 m

False northing: 10,000,000 m

Scale factor: 0.9996

Why NZTM is ideal for NZ:

Conformal — preserves local angles and shapes

Metre-based — direct distance/area measurement

Centred on NZ — minimal distortion across the country

Standard — used by LINZ, councils, and all major consultancies

Christchurch in NZTM:

E: 1,571,747 m N: 5,180,043 m

LIVE DEMO

[L04_CRS_Explorer.html](#)

See how the same geographic features look different under different coordinate reference systems and projections.

Scan to open the demo on your phone



aardid.github.io/uc_205_gis_course/L04_CRS_Explorer.html

Vertical Datums — The Third Dimension

Horizontal CRS tells you WHERE (lat/lon or easting/northing)

Vertical datum tells you HOW HIGH (elevation above what?)

NZVD2016 — NZ Vertical Datum 2016

National standard — heights relative to the geoid (NZGeoid2016)

Replaces 13 different local vertical datums

Lyttelton Vertical Datum 1937 (LVD-37)

Historical Canterbury datum — based on Lyttelton tide gauge

Still used in some older datasets and plans

Christchurch Drainage Datum (CDD)

Local datum for stormwater and wastewater design — offset from LVD-37

Mixing vertical datums = wrong elevations = pipes that flow the wrong way!

Always ask: 'heights relative to what?'

CRS in Python — GeoPandas

Checking CRS:

```
gdf.crs          # returns the CRS object  
gdf.crs.to_epsg() # returns EPSG code (e.g. 2193)
```

Setting CRS (declaring, no transformation):

```
gdf = gdf.set_crs(epsg=4326) # label as WGS84  
gdf = gdf.set_crs('EPSG:4326') # same thing
```

Reprojecting (transforms coordinates):

```
gdf = gdf.to_crs(epsg=2193) # reproject to NZTM  
gdf = gdf.to_crs('EPSG:2193') # same thing
```

Golden rule: check → set if missing → reproject to EPSG:2193 → then analyse

set_crs() vs to_crs() — The Most Common Mistake in GIS

They sound similar, but do very different things.



set_crs() — Declare (Label)

Assigns a CRS label to data that is already in that coordinate system.
It does **NOT** change the numbers.

⚠ Before

CRS = None

(172.6, -43.5)

set_crs(4326)



✅ After

CRS = EPSG:4326

(172.6, -43.5)



Coordinates DO NOT change!

set_crs() just LABELS the data — it tells Python what the numbers already mean.



Analogy

Like writing “kg” on a box that was already weighed in kilograms — the weight doesn’t change.



to_crs() — Transform (Reproject)

Transforms coordinates from one CRS to another.
It **DOES** change the numbers.

🌐 Before

CRS = EPSG:4326

(172.6, -43.5)

to_crs(2193)



✅ After

CRS = EPSG:2193

(1571747, 5180043)



Coordinates CHANGE!

to_crs() TRANSFORMS the coordinates — mathematically reprojects every point to the new system.



Analogy

Like converting 10 kg to 22.05 lbs — the quantity is the same, but the numbers change.



Key takeaway

- Use set_crs() when you are declaring what CRS your data is already in (no numbers change).
- Use to_crs() when you need to convert your data to a different CRS (numbers change).

Rule of thumb

Label the data with set_crs().
Move the data with to_crs().

Confusing these two is the #1 CRS mistake in GIS. If in doubt: to_crs() is the one that moves things. Always ask: 'am I labelling or converting?'

CRS Best Practices for NZ Engineering

1. Always check CRS first — `gdf.crs` — before any analysis
2. Use `set_crs()` only when CRS is None (missing metadata)
3. Use `to_crs()` to reproject — this transforms coordinates
4. Reproject all layers to the same CRS before overlay or join
5. Use EPSG:2193 (NZTM) for distance, area, and buffer calculations in NZ
6. Keep original data untouched — reproject into a new variable
7. Document which CRS you used in your analysis

If in doubt: EPSG:2193 is almost always the right choice for NZ work.

Today's Takeaways

1. The Earth has three surfaces: physical, geoid, ellipsoid — a datum positions the ellipsoid
2. Geographic CRS uses degrees (lat/lon) — not suitable for distance/area calculations
3. Projected CRS uses metres — every projection distorts something (area, shape, distance, direction)
4. In NZ: EPSG:4326 (geographic) and EPSG:2193 / NZTM2000 (projected) are the two you need
5. `set_crs()` labels, `to_crs()` transforms — confusing them is the #1 CRS mistake

Consequences of Wrong CRS

Overlay Errors

Layers don't align — features appear in the ocean or shifted by hundreds of km

Area Miscalculation

`gdf.area` in degrees gives numbers like 0.0001 — meaningless for engineering

Distance Errors

Buffers in degrees are ellipses, not circles — wrong in both shape and size

Silent Failures

Python won't crash — it will happily give you wrong answers



Imagine this coastline shifted 100s of metres because of a CRS mismatch.

Always ask: 'what CRS is this in?'

Canterbury Case Study -- Post-Earthquake Datum Shifts

The 2010–2011 earthquakes physically moved Canterbury.

Horizontal shifts: up to 4–5 metres in some areas

Vertical changes: up to 1.5 m subsidence in eastern suburbs, 0.5 m uplift elsewhere

Impact on CRS:

Pre-quake survey marks no longer matched their published coordinates

NZGD2000 coordinates were updated — 'deformation model' patches applied

Old plans and datasets became inconsistent with post-quake reality

Vertical datums shifted — drainage designs based on pre-quake levels failed

Lesson: CRS is not static -- it can change when the Earth moves. Always check the epoch/version.

Always ask: 'when was this surveyed?'

**A colleague sends you coordinates (172.6, -43.5).
What CRS are they in? How do you verify?**

Think about:

Are these degrees or metres? How can you tell from the numbers alone?

If they're degrees, what CRS? WGS84 or NZGD2000?

What would you need to do before using these in a distance calculation?

What if the colleague sent (1571747, 5180043) instead?